Morris Greenberg

Ph.D. Candidate in Statistical Sciences

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Education

PhD **Department of Statistical Sciences, University of Toronto**, Statistical Sciences — Toronto, ON, Canada, Sept 2021 – present

- Advisors: Prof. Radu Craiu, Prof. Kieran Campbell
- Research interests: Bayesian modeling and programming, scalable methods for complex health-care data (especially ‘omics data), graph inference, hierarchical models, approximate Bayesian computation

MS **Department of Statistical Science, Duke University**, Statistical Science — Durham, NC, USA, Aug 2019 – May 2021

BS **Tufts University**, Quantitative Economics and Mathematics — Medford, MA, USA, Aug 2012 – May 2016

Awards and Honors

NSERC Postgraduate Scholarship - Doctoral — Canada
$101,000 CAD, National Sciences and Engineering Research Council — Sept 2023 – Aug 2026

Queen Elizabeth II Graduate Scholarship in Science and Technology — Ontario
$15,000 CAD, Ontario Student Assistance Program — Sept 2022 – Aug 2023

Doctoral Recruitment Award — Toronto, ON
$5,000 CAD, University of Toronto — Sept 2021

BEST Award for Master’s Research in Statistical Science — Durham, NC
$500 USD, Duke University — June 2021

Conference Presentations

- Greenberg, M., Campbell, K., Craiu, R. (2026, June). Estimating Hierarchical Tissue Organization with Structure Learning. 2026 International Society for Bayesian Analysis World Meeting, Nagoya, Japan. (Poster)
- Greenberg, M., Campbell, K., Craiu, R. (2025, September). Restricted Search Space Graph MCMC via Birth-Death Processes. The Fast and Curious 2: MCMC in Action, Toronto, ON, Canada.
- Greenberg, M., Campbell, K., Craiu, R. (2025, May). Restricted Search Space Graph MCMC via Birth-Death Processes. Statistical Society of Canada Annual Meeting 2025, Saskatoon, SK, Canada. Advances in Statistical Inference and Computing.
- Greenberg, M., Campbell, K., Craiu, R. (2025, January). Restricted Search Space Graph MCMC via Birth-Death Processes. University of Florida Department of Statistics' Annual Winter Workshop: 2025 Computational Methods in Bayesian Statistics, Gainesville, FL, USA. (Poster)
- Greenberg, M., Craiu, R., Campbell, K. (2024, June). Restricted Search Space Graph MCMC via Birth-Death Processes. International Chinese Statistics Association Canada Chapter Symposium, Niagara Falls, ON, Canada. Advances in Monte Carlo Methods.
- Greenberg, M., Craiu, R., Campbell, K. (2023, May). Restricted Search Space MCMC With Adaptive Weighting And Sparsity Parameterization For Graph Inference. Statistics Society of Canada Meeting 2023, Ottawa, ON, Canada. Advances In Bayesian Modelling And Computation.
- Greenberg, M., Craiu, R., Campbell, K. (2023, May). Restricted Search Space MCMC Methods for Graph Inference. The Fast and Curious: Modern Markov Chain Monte Carlo, Minneapolis, MN, USA.
- Greenberg, M., Ma, L., Wang, Z., Ma, P., Sung, A. (2021, August). Logistic Tree Gaussian Processes (LoTGaP) for the Microbiome. Joint Statistical Meetings 2021, Next-Generation Sequencing and High-Dimensional Data Section.

Preprints and Publications

Restricted Search Space Graph MCMC via Birth-Death Processes 2026
Morris Greenberg, Kieran R. Campbell, Radu V. Craiu
(arXiv:2604.10863 [stat])

Chlorhexidine Gluconate Bathing Reduces the Incidence of Bloodstream Infections in Adults Undergoing Inpatient Hematopoietic Cell Transplantation Jan 2021
Vinay K. Giri, et al.
10.1016/j.jtct.2021.01.004 (Transplantation and Cellular Therapy)

Development of predictive risk models for major adverse cardiovascular events among patients with type 2 diabetes mellitus using health insurance claims data 2018
J.B. Young, M. Gauthier-Loiselle, R.A. Bailey, A.M. Manceur, P. Lefebvre, Morris Greenberg, M.H. Lafeuille, M.S. Duh, B. Bookhart, C.H. Wysham
(Cardiovascular Diabetology, 17(1), p.118)

Development of Risk Models for Major Adverse Chronic Renal Outcomes Among Patients with Type 2 Diabetes Mellitus Using Insurance Claims - A Retrospective Observational Study 2019
C.H. Wysham, M. Gauthier-Loiselle, R.A. Bailey, A.M. Manceur, P. Lefebvre, Morris Greenberg, M.S. Duh, J.B. Young
(Current Medical Research and Opinion)

Work Experience

University of Toronto Statistical Sciences Department, Course Instructor — Toronto, ON
Jan 2025 – May 2025
- Course co-instructor for An Introduction to Statistical Reasoning and Data Science (STA 130)

University of Toronto Statistical Sciences Department, Teaching Assistant — Toronto, ON — Aug 2021 – present

- Course list includes: Advanced Computational Methods for Statistics I (STA 2311, Fall 2024), Bayesian Statistics (STA 365, Winters 2022-2025, Fall 2025), Statistical Consultation, Communication, and Collaboration (STA 490, 2022-2025), Introduction to Machine Learning (STA 314, Fall 2022, Winter 2026), Data Science in Practice (STA 2546, Winter 2022)

Li Ma Statistics Lab / Duke University Hospital, Research Assistant — Durham, NC — May 2020 – Aug 2021

- Developed novel hierarchical models using Gaussian processes to analyze the dynamics of microbiome composition of patients in bone marrow transplant studies
- Designed a microbiome logistic tree package in R that includes microbiome methods for mixed effects modeling, Gaussian process modeling, and topic modeling

Analysis Group, Analyst / Senior Analyst — Boston, MA — Aug 2016 – May 2019

- Developed models to predict cardiovascular risk, chronic kidney disease, and chronic thromboembolic pulmonary hypertension (CTEPH) in Type II Diabetes Mellitus patients, which resulted in multiple academic journal articles
- Created an elasticity model to describe the price sensitivity of patients, physicians, and providers towards cholesterol lowering drugs in expert reports for intellectual property litigation
- Built machine learning tools (e.g. neural networks, random forests, Gaussian mixture models) to identify user types of opioids for the purposes of targeted treatment care
- Analyzed the source code for millions of files across programming languages for tax litigation; measured the similarity between pairs of files to understand how source code was modified by developers over time

Washington Nationals, Baseball Operations Intern / Consultant — Washington, DC — May 2015 – Apr 2016

- Derived, coded, and maintained a Markov Chain simulator for the team
- Worked on analytical problems using statistical learning methods (e.g. random forests, gradient boosted algorithms, and generalized linear models) in R

Skills

Programming: R, Python, C++, STAN, Matlab, LaTeX, Stata, SAS, SQL (MySQL, Oracle SQL Developer)

Languages: English, Spanish, Hebrew

Miscellaneous: Top 100 Scrabble Player in North America; came in 10th at the 2017 North American Championships